WASP-11b

R-0797

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-10_221113` · created 2026-07-28T08:27:33+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459896.7056 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.051
Transit depth (Rp/Rs)^2	3.12 +/- 1.8 [%]
Orbital Inclination (inc)	89.74 +/- 1.72
Airmass coefficient 1 (a1)	0.27 +/- 0.19
Airmass coefficient 2 (a2)	0.69 +/- 0.83
Scatter in the residuals of the lightcurve fit is	71.42 %
Best Comparison Star	#1 - [582, 275]
Optimal Aperture	2.53
Optimal Annulus	5.75
Transit Duration (day)	0.106 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.177 ± 0.051 vs 0.131 (deviation 0.9σ)
Duration fit vs published	0.106 d vs 0.1075 d (deviation 0.15σ)
Mid-time O–C	-4.4 min
Depth SNR	3.5
Residual scatter	71.42 %

Light curve

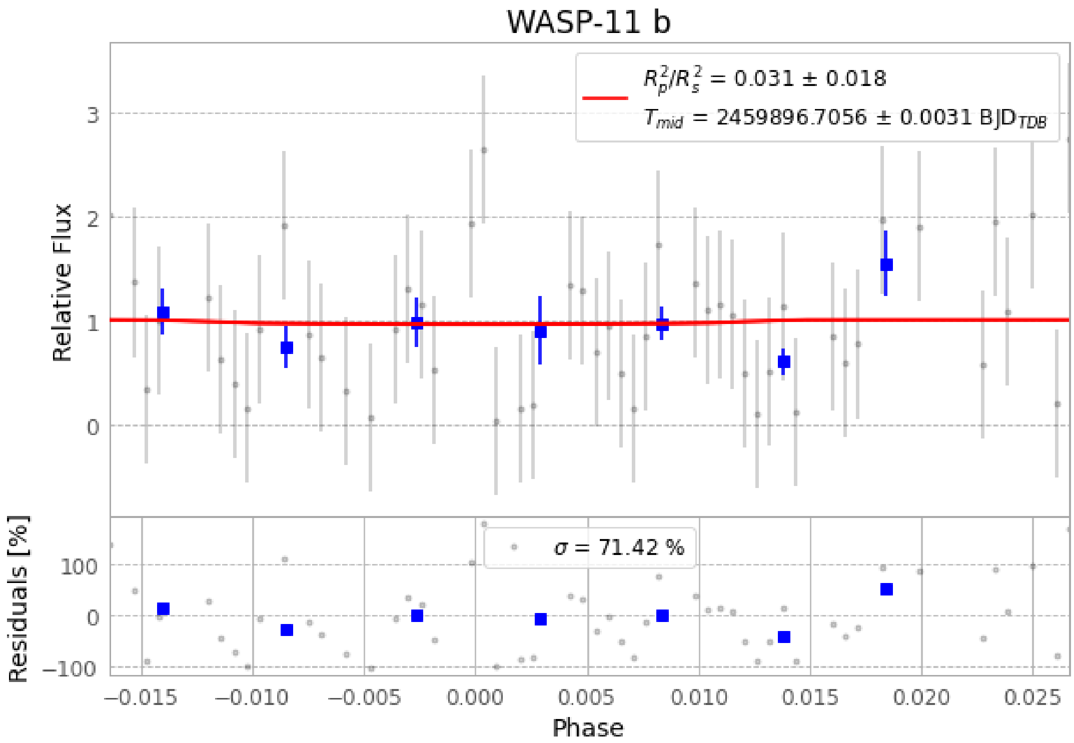

Figure 1 — FinalLightCurve_WASP-11 b_2022-11-12.png (SHA-256 38ef6e50d6f09ea6…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [500, 354], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 7ef5c07a9156b81e…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

78 FITS frames (51 MB), `HATP-10221113031815.FITS` → `HATP-10221113070915.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-221113.log (3b175acd9e79…), FinalLightCurve_WASP-11 b_2022-11-12.png (38ef6e50d6f0…), FinalLightCurve_WASP-11 b_2022-11-12.csv (c60c19820e66…)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)